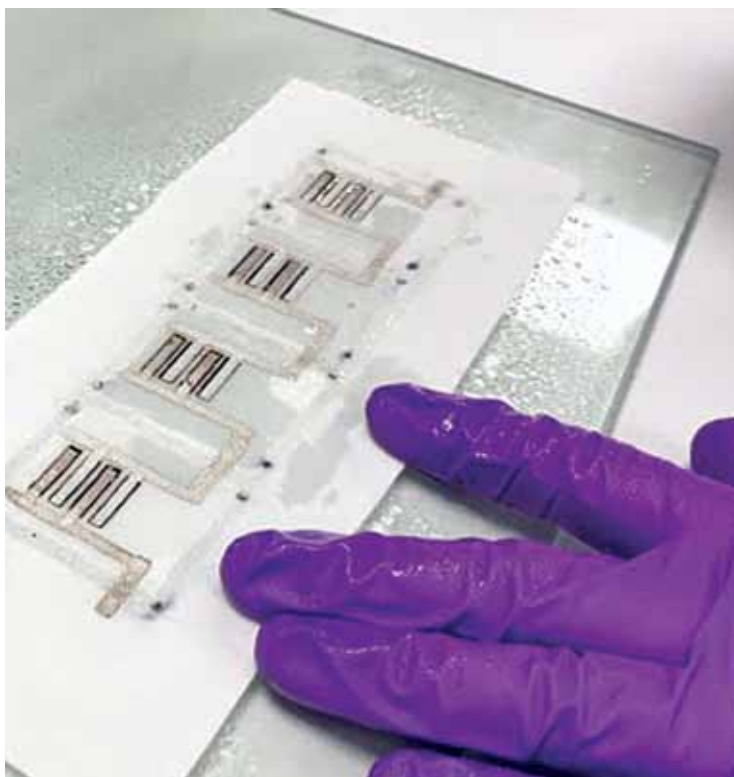


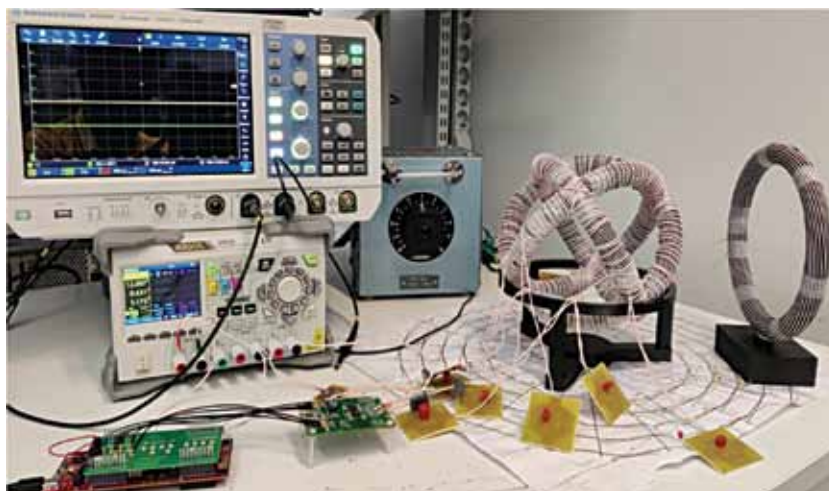
Sweat-powered battery for wearable technologies

Scientists from Nanyang Technological University have developed a soft and stretchable battery that is powered by human perspiration. The prototype battery consists of printed silver flake electrodes that generate electricity in the presence of sweat. Measuring 2cm x 2cm, and as flat as a small paper bandage, the battery is affixed to a flexible and sweat absorbent textile that is stretchable and attachable to wearable devices like watches, wristbands, or arm straps. An individual wearing the battery around their wrist can generate a voltage of 4.2V and output power of 3.9mW, sufficient to power a commercial temperature sensor device and send the data continuously to a smartphone via Bluetooth. The battery could help reduce harmful electronic waste; it does not contain heavy metals or toxic chemicals.

*It is about the size of a thumbnail and can be attached to wearable devices
(Credit: www.ntu.edu.sg)*



Wireless charging technology for charging multiple devices simultaneously



Components of the device (Credit: www.aalto.fi/en)

Researchers across the globe are trying to make truly wireless charging solutions, but they all employ sophisticated technologies that involve detecting the presence and position of the phone or charging device for sending energy in its direction, usually with cameras or sensors, adding bulk and cost to the device. Now researchers from the Aalto University have developed a new wireless transmitter that bypasses the need for a detection sensor and transfers power channels in all directions, automatically tuning channels when receiving devices are in motion. Currently, the transmitter is highly efficient at ninety percent at up to twenty centimetres distance, but can also work at longer distances, just with a lower efficiency of energy transfer.

High energy density micro batteries

With portable electronic devices becoming more compact in size, their batteries are required to store more power in less space. But the energy density gets harder to improve with a small form factor, as a large portion of the battery's

footprint must be devoted to protective packaging. Therefore, researchers from the University of Pennsylvania's School of Engineering and Applied Science have shown a new way to build and package micro batteries that maximises energy